MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF CIVIL ENGINEERING			
END SEMESTER EXAMINATION (MAKEUP)		**December, 2024**	
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**Civil Engineering**	**Course Type**	**FC**
Course Name	**Precast Technology**	**Course Code**	**CIE 3125**
Semester	**V**	**Max Marks**	**50**
Date		**Duration**	**3 hrs**

Instructions:

- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		BL
			CO	PO	
1A	Classify the large panel wall system based on i) structural functions ii) construction and iii) location and functional requirements.	**4**	3	1,2,4,5,7,10,11,PSO2	**2**
1B	Describe the characteristics and application of 2D precast systems.	**3**	3	1,2,4,5,7,10,11,PSO2	**2**
1C	Explain the process of pretensioning as a method of prestressing. List the advantages and limitations.	**3**	2	1,4,7,12,PSO2	**2**
2A	With the help of a neat sketch, illustrate the fire safety consideration to be adopted for passive fire protection in buildings.	**4**	3	1,2,4,5,7,10,11,PSO2	**2**
2B	With the help of a neat sketch, illustrate the major types of precast slab elements. Identify the suitability of each type in precast construction.	**4**	3	1,2,4,5,7,10,11,PSO2	**2**
2C	Describe the technological breakthrough during the developing years that led to the demonstration of precast concrete during 1920-1940.	**2**	1	1,11,12,PSO2	**2**
3A	With the help of a neat sketch, illustrate the parts of precast pile and precast open foundation.	**4**	3	1,2,4,5,7,10,11,PSO2	**2**

3B	Identify and explain the suitability of equipment required for erection of precast components.	**3**	4	1,4,5,6,9,11,12,PSO2	**2**
3C	Classify the various types of equipment used in the precast concrete industry/construction.	**3**	4	1,4,5,6,9,11,12,PSO2	**2**
4A	Identify the work involved in the manufacture of precast elements under auxillary and subsidiary process.	**4**	4	1,4,5,6,9,11,12,PSO2	**2**
4B	Illustrate the plant process involved for producing the structural precast elements.	**3**	4	1,4,5,6,9,11,12,PSO2	**2**
4C	In a project involving the installation of precast wall panels for a commercial building, following defects were observed: Chip-off at panel edges due to excessive force exerted on the elements when handling. Propose the remedial measures to ensure the longevity and performance of the wall panels.	**3**	4	1,4,5,6,9,11,12,PSO2	**2**
5A	Explain the following in the context of the design of the joints for connecting various precast elements: a) serviceability b) appearance and c) water tightness and d) practicability	**4**	5	1,2,11,12,PSO2	**2**
5B	Identify the types of connections and jointing techniques in precast construction.	**3**	5	1,2,11,12,PSO2	**2**
5C	Describe the suitability of battery form method and tilting mould method for manufacture of precast elements.	**3**	4	1,4,5,6,9,11,12,PSO2	**2**

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO	Total Marks
1	2
2	3
3	19
4	19
5	7

BL	Total Marks
2	**50**
3	
4	
5	